Dockstore - 2021 update

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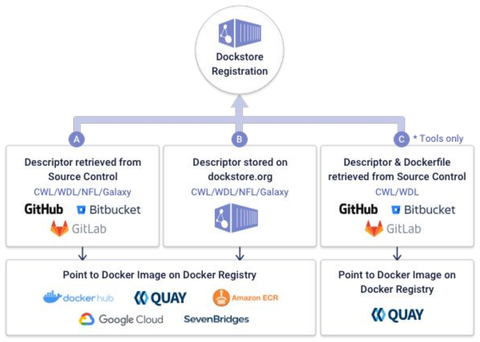
Background

Dockstore is a free and open source platform for sharing scientific tools and workflows. It is primarily a registry of Docker-based resources described using popular workflow languages in bioinformatics such as CWL, WDL, Nextflow, and now Galaxy.

Our initial aims were to increase portability (sharing workflows that can run in any environment supporting Docker), interoperability (championing and implementing GA4GH APIs), and reproducibility (preserving dependencies via Docker and describing workflows with popular descriptor languages)

We are now on version 1.11 after last presenting version 1.6.0 at BOSC 2019.

Learn more from our training materials at https://docs.dockstore.org/en/stable/ and our recent paper at https://doi.org/10.1093/nar/gkab346



Caption 1: Ways of getting workflows into Dockstore

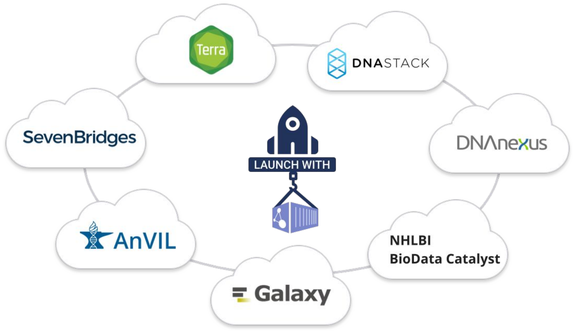
Automatic updates

Traditionally, workflow developers login to Dockstore and update their entries. With GitHub apps, they can automatically update their repos with changes to GitHub and also control:

- what branches and and tags make it to Dockstore (more useful for repos with a complex branching structure)
- control publishing
- control default branch (useful when automatically updating the branch presented to other users)

Highlighted New Features

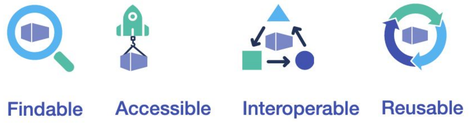
- Added launch-with support for
  - multiple instances of Galaxy
  - CGC
  - ANVIL
  - BD Catalyst
  - Cavatica
  - Galaxy



Caption 2: Depiction of launch with partners

- Usability improvements
  - Redesigned logged-off homepage and about page
  - Redesigned logged-in homepage
    - Easily access your recently modified tools and workflows
    - Monitor your starred recently modified tools, workflows, and organizations
- Behind the scenes
  - Language plugins allow for contributed code to accelerate new languages in Dockstore (for validation, parsing, visualization, etc.)
  - Performance improvements for large repos with many branches, viewing your own workflows across many orgs, pagination
  - Get excited if things like FedRAMP or CIV AWS standards mean something to you
  - Updates
    - WDL 1.0
    - CWL 1.1
    - Nextflow DSL2
    - Swagger 2.0 to OpenAPI 3.0 for our API
- More details at https://github.com/dockstore/dockstore/releases/tag/1.11.5 and past releases

FAIR work on Dockstore



What is FAIR?

- A set of principles that encourage **F**indable, **A**ccessible, **I**nteroperable, and **R**eusable practices

How do you create FAIR workflows?

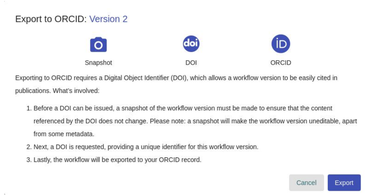
- Follow our new Best Practices documentation

How do you find FAIR work on Dockstore?

- "Star" Feature:
  If an entry or an organization has been "starred", their recent activity appears on the user's homepage
- Organizations/Collections Feature:
  Users can find more information about the members of an organization or contributors to a collection through the members tab. ORCID ID can also be displayed, thereby connecting users with the members' scientific contributions.

Snapshot, DOI, and ORCID

A new wizard walks users through snapshotting workflow versions, requesting DOIs for them in Zenodo, and introduces being able to export them to your ORCID profile



Caption 3: Wizard for ORCID export

Learning Resources

Dockstore fundamentals: Video tutorials

Dockstore documentation

Notable work on Dockstore

- Benchmarking Information: The vcf-to-gds workflow has detailed performance benchmarking information relating the size of the input samples to processing time, cost and other workflow configuration related parameters
- nf-core has set up an organization of automatically updated workflows on Dockstore and has best practice documents at https://nf-co.re/ for Nextflow workflows
- Galaxy has established IWC (Galaxy Intergalactic Workflow Commission) with best practice workflows
- Dockstore also provides recommendations for best practices when working with reproducible workflows in Dockstore

Future Work

How can a user know if a workflow on Dockstore is useful, reproducible, reusable, and follows secure best practices? We are working on the following new features to help measure these and build trust:

- Workflow Accreditation where organizations can give their "stamp of approval"
- "Report Cards" that display reproducibility measures associated with workflows
- FedRAMP authorization
  - We are building systems and procedures to improve security of our software, which will be routinely audited by third parties for compliance

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